Tobias Mwangi-Hartley

PROFESSIONAL PROFILE

Data Scientist with experience building and evaluating forecasting, churn-modelling and experimentation solutions using Python, SQL and cloud data tools. Work includes feature engineering, scheduled model workflows, batch scoring and measurable business outcomes. Transitioning toward machine learning engineering, with recent experience adding pytest unit tests to forecasting feature code and building a Dockerfile for a Streamlit application.

TECHNICAL SKILLS

Programming and data: Python, SQL, R, pandas, scikit-learn

Machine learning and statistics: LightGBM, XGBoost, SHAP, statsmodels, feature engineering, forecasting, churn modelling, experimentation, Bayesian hierarchical modelling

Platforms and tools: BigQuery, Vertex AI Workbench, Looker, dbt, dbt Cloud scheduler, Streamlit, Docker, pytest, R Markdown

Learning project: PyTorch, torchvision, Google Colab

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox | 2022–Present

- Built a weekly recipe-demand forecasting workflow using LightGBM, with calendar, price and menu-position features to forecast orders by recipe and region six weeks ahead.
- Scheduled the forecasting notebook in Vertex AI Workbench to write results to BigQuery for operational use in Looker. Improved WAPE from 23% to 14%; estimated food-waste reduction was 9%.
- Developed a weekly XGBoost churn-scoring workflow in BigQuery, using dbt, and supported retention targeting of top-decile customers. In a 10% holdout validation, churn was 2.1 percentage points lower in the treated group.
- Built a Streamlit calculator for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Wrote its Dockerfile; the data platform team handled deployment to Cloud Run.
- Began writing pytest unit tests for forecasting feature code in 2025; coverage is approximately 60%.
- Delivered the internal training session “Stats for PMs” three times.

Junior Data Analyst | Firth Energy Retail | 2020–2022

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

EDUCATION

MSc Statistics

- Dissertation: Bayesian hierarchical models

ADDITIONAL PROJECT

Image Classification Course Project | Personal | 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset as part of an online deep-learning course; achieved approximately 80% accuracy.
- Learning project using PyTorch, torchvision and Google Colab.